

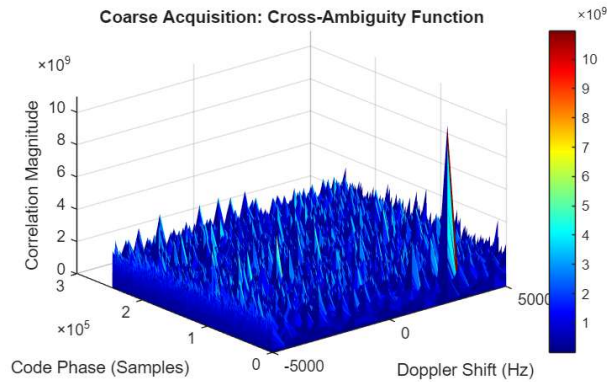
[Part 1] Signal Generation & Cryptographic Initialization

[Init] Generating ECDSA (secp256r1) Key Pair...

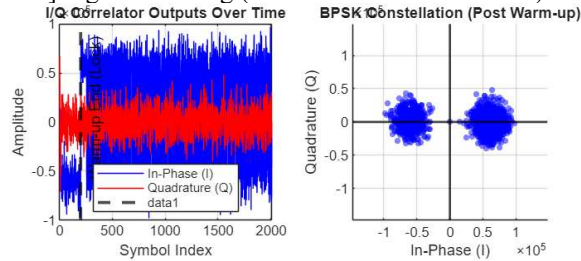
[Tx] Generating TESLA Hash Chain (n=10)...

[Tx] Deterministic puncturing and Payload Insertion.

[Part 2] Coarse Signal Acquisition



[Part 3] Signal Tracking (Realistic PLL + Ideal DLL)



[Part 4] CNAV-2 Data Channel Decoding (BCH & LDPC)

[Rx] Subframe 1 TOI (BCH) Recovered: 0

[Rx] Subframe 2 (LDPC): CRC Passed!

[Rx] Subframe 3 (LDPC): CRC Passed!

[Rx] Receiver successfully reconstructed the pure raw subframes (926 bits).

[Part 5] Physical Payload Extraction & Viterbi Decoding

[Rx] Viterbi Decoding Errors: 0 / 900

[Part 6] Cryptographic Verification (Receiver Side)

[Rx] Pilot CRC Check : PASS

[Rx] TESLA Key Chain Verify : PASS (Hash matched stored commitment K_3)

[Rx] Root Signature Verify : PASS (ECDSA-512 Validates Derived K_0)

[Rx] Navigation MAC Verify : PASS (Navigation Subframes are Authentic & Intact)

[+++] Cryptographic Authentication Fully Successful! Signal is Genuine. [+++]

[Part 7] CNAV-2 Navigation Message Parsing (SF1, SF2, SF3) ===

[Rx] Subframe 1 (BCH Decoded)

- Time of Interval(TOI): 0

[Rx] Subframe 2 (LDPC Decoded - Clock & Ephemeris)

- Week Number (WN) : 2149

- Interval TOW (ITOW) : 0

- L1C Health Flag : 0 (0 = Healthy)

- t_{oe} (Ref Time) : 0 [sec]

- Eccentricity (e) : 0.020000

[Rx] Subframe 3 (LDPC Decoded - Variable Page Data)

- PRN ID : 7

- Page Number : 4

- Page Contents : Midi Almanac
